UGANDA REVISION HUB

S4 Starter Study Notes

Batch 1 • Online and offline revision

S4 • Mathematics

Statistics: mean, median, mode and range

Learning objectives

- Calculate the mean of a data set
- Find the median and mode
- Calculate the range and interpret results

Lesson notes

The mean is the sum of all observations divided by the number of observations. The median is the middle value after arranging the data in order. The mode is the value that occurs most often.

The range is the largest value minus the smallest value. It gives a simple measure of spread.

For an even number of observations, the median is the mean of the two middle values after arranging the data.

Worked example

For 2, 3, 3, 6, 11: mean = 25/5 = 5; median = 3; mode = 3; range = 11 - 2 = 9.

Revision questions

1. Find the mean of 4, 6, 8 and 10.
2. Find the median of 1, 3, 7, 9, 12.
3. Find the mode and range of 5, 5, 6, 8, 10.

Suggested answers

1. 7
2. 7
3. Mode = 5; range = 10 - 5 = 5

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S4 • Biology

Reproduction in flowering plants

Learning objectives

- Identify parts of a flower involved in reproduction
- Distinguish pollination from fertilisation
- Explain how seeds and fruits form

Lesson notes

The stamen is the male reproductive part of a flower and includes anther and filament. The carpel or pistil is the female part and includes stigma, style and ovary.

Pollination is the transfer of pollen grains from anther to stigma. Fertilisation occurs when the male nucleus from a pollen grain fuses with the female nucleus in an ovule.

After fertilisation, the ovule develops into a seed and the ovary usually develops into a fruit. Pollination may be carried out by insects, wind or other agents.

Worked example

A bee may carry pollen from the anther of one flower to the stigma of another flower of the same species.

Revision questions

1 Define pollination.

2 Which flower part contains ovules?

3 State one difference between pollination and fertilisation.

Suggested answers

1 Transfer of pollen from anther to stigma

2 Ovary

3 Pollination is pollen transfer; fertilisation is fusion of male and female nuclei.

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